A PROJECT REPORT

on

"QR-Based Digital Library System (LIBRIX)"

Submitted by

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Seat no. 26CS510085

in partial fulfilment for the award of degree of

BACHELOR OF SCIENCE

in

COMPUTER SCIENCE

Under the guidance of

Prof. Shinoj Mathew

DEPARTMENT OF COMPUTER SCIENCE



MAHATMA EDUCATION SOCIETY'S
PILLAI HOC COLLEGE OF ARTS, SCIENCE & COMMERCE,
(Autonomous) RASAYANI, 410207 MAHARASHTRA
(Affiliated to Mumbai University)
NAAC Accredited with A+ Grade in Cycle2

Sem V / VI
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DEPARTMENT OF COMPUTER SCIENCE

CERTIFICATE

This is to certify that **Ms. Neha Vinod Mhatre** of **TY BSc (SEM V / VI)** class has satisfactorily completed the Project "**QR-Based Digital Library System (LIBRIX)"** submitted in the partial fulfilment of award of **Bachelor of Science** in **Computer Science** during the academic year **2026-2027**

Date of submission:

Project Guide

Head / In charge,
Department of Computer Science

College seal

Signature of Examiner

DECLARATION

I **Neha Vinod Mhatre**, hereby declare that the project entitled, **"QR-Based Digital Library System (LIBRIX)"** submitted in the partial fulfilment of award of "**Bachelor of Science** in **Computer Science"** during the academic year **2026-2027** is my original work and the project has not formed the basis for the award of any degree, associateship, fellowship or any other similar titles.

Signature of the Student**:**

Name: **Neha Vinod Mhatre**

Place: **Rasayani**

Date:

ACKNOWLEDGEMENT

It gives me great pleasure to present this project report on **— "QR-Based Digital Library System (LIBRIX)"** It's a great pleasure and moment of immense satisfaction for me to express my profound gratitude to my Project Guide, **Shinoj Mathew,** whose constant encouragement enabled me to work enthusiastically. Her perpetual motivation, patience and excellent expertise in discussion during progress of the project report work have benefited me to an extent, which is beyond expression. I am highly indebted to her invaluable guidance and ever-ready support in the successful completion of this project report in time. Working under her guidance has been a fruitful and unforgettable experience. I would like to thank all people who contributed to the successful completion of the project.

I would like to thank our principal **Principal Dr. Rinko Shantnu** & Department **Coordinator Ms. Priyanka Sorte,** for their support.

- Neha Vinod Mhatre

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CHAPTER 1:

INTRODUCTION

1.1 Background

A library contains a large amount of information, but managing books manually can become difficult as the number of books and users increases. The **QR-Based Digital Library System (LIBRIX)** is a web-based library management application developed using Python and Flask. It provides a digital way to register users, log in, search books, check availability, issue books, return books, and manage library records. The project also generates a QR code for each book so that every book record has a digital identification reference.

The supplied project files show Flask for the web application, Flask-SQL Alchemy for database operations, SQLite for local storage, Werk Zeug security functions for password hashing and verification, and the QR-code package for generating book QR codes. The supplied CSS defines LIBRIX home, authentication, dashboard, and responsive layouts.

1.2 Motivation

- Reduce the effort required to maintain library records manually.
- Make book searching and availability checking faster for students.
- Keep issue and return information in a central database.
- Provide separate facilities for students and administrators.
- Give each book a QR-code reference for easier digital identification.
- Provide a clean and responsive interface.

1.3 Problem Statement

Traditional library management may depend on registers or separate manual records. Searching, checking availability, recording issue dates, and tracking returns can become slow as the collection grows. LIBRIX addresses this problem by providing one digital system where students can search and borrow books while administrators manage books, users, and issue records.

1.4 Objectives

- Develop a web-based digital library using Flask and SQLite.
- Allow students to register and log in.
- Search books by title, author, category, or ISBN and filter by availability.
- Issue available books and automatically calculate the expected return date.
- View and return issued books.
- Provide administrator functions for users, issues, and books.
- Generate QR-code images for book records.
- Store library information in structured relational tables.
-

1.5 Scope of the Project

The project covers registration, login, student dashboard functions, book browsing and search, book details, issue and return processing, administrator functions, QR-code generation, and SQLite storage. A dedicated camera-based QR scanning workflow is not evident in the supplied application code, so this report describes the implemented QR feature as QR generation and book identification rather than claiming a completed scanner module.

CHAPTER 2:

LITERATURE SURVEY / EXISTING SYSTEM

2.1 Existing System

A conventional library generally uses a catalogue and librarian-assisted issue and return records. A student searches for a book, checks availability, requests it, and the transaction is recorded. This works for small collections but becomes time-consuming when many books and users are involved.

2.2 Limitations of Existing System

- Searching can take more time when records are maintained manually.
- Availability information may not be updated immediately.
- Issue and return records require repeated data entry.
- Finding an old transaction in a large register is difficult.
- Student and administrator activities are less clearly separated.
- Manual book identification can be slower than using a digital identifier.

2.3 Related Work and Technology Direction

Digital library systems commonly use a database-backed catalogue, authentication, search, and transaction management. LIBRIX follows this model with a lightweight Flask application and adds QR-code generation for individual book records. The supplied CSS also provides responsive layouts for home, authentication, and dashboard pages.

2.4 Need for the Proposed System

A small library needs a system that is simple to operate but structured enough to maintain reliable records. LIBRIX keeps users, books, and issue transactions in related database tables and applies role-based access to separate student and administrator operations.

CHAPTER 3 :

METHODOLOGY

3.1 Overview

LIBRIX is a centralized web-based digital library. After login, a student can use the dashboard, browse books, search the catalogue, view book details, issue available books, and view or return issued books. An administrator has additional access to user, issue, and book management.

3.2 Core Functional Modules

3.2.1 Authentication Module

Registration collects name, email, and password. The application checks required fields, validates password length, prevents duplicate email registration, and verifies passwords during login. The user ID and role are kept in the session after successful login.

3.2.2 Student Module

- Dashboard and library information.
- Book browsing, search, suggestions, and details.
- Issue an available book for a 14-day period.
- View personal issued-book records.
- Return an issued book.

3.2.3 Administrator Module

- Administrator dashboard.
- User management and role changes.
- Issue-record management.
- Administrator book returns.
- Add books with title, author, ISBN, category, shelf, and quantity.
- Delete books when they are not currently issued.

3.2.4 QR Code Module

When a book is added, the application creates a QR-code image in the static QR-code directory and stores the QR-code path in the book record. A separate routine can regenerate QR codes for existing books.

3.3 Advantages of Proposed System

- Faster catalogue searching.
- Centralized records.
- Separate student and administrator access.
- Automatic availability updates.
- Automatic due-date calculation.
- QR-code generation for books.
- Responsive web interface.

3.4 Feasibility Study

Technical: The project uses common Python web technologies and a local SQLite database.
Economic: The core stack is lightweight and suitable for student development.
Operational: The interface is based on familiar actions such as login, search, issue, return, and administration.

CHAPTER 4:

WORK DESCRIPTION, OBSERVATION & ANALYSIS

System Requirements

1. Hardware Requirements

Component	Requirement
Processor	Intel i3 / equivalent or above
RAM	4 GB minimum; 8 GB suggested
Storage	At least 2 GB free
Display	Standard laptop/desktop display
Network	Local network or internet as required

2.Software Requirements

Software / Technology	Purpose
Python	Application programming language
Flask	Web framework
Flask-SQL Alchemy	Database ORM integration
SQLite	Local relational database
Werk Zeug Security	Password hashing and verification
QR-code	QR-code image generation
HTML / Jinja	Web page templates
CSS	Interface styling and responsive design
Web Browser	Application access

3. Database Requirements

The supplied SQLite database contains three main tables: **user**, **book**, and **issue**. The user table stores identity and role information. The book table stores catalogue and QR information. The issue table connects users and books and stores issue, due, return, and status information.

4.Current Database Snapshot

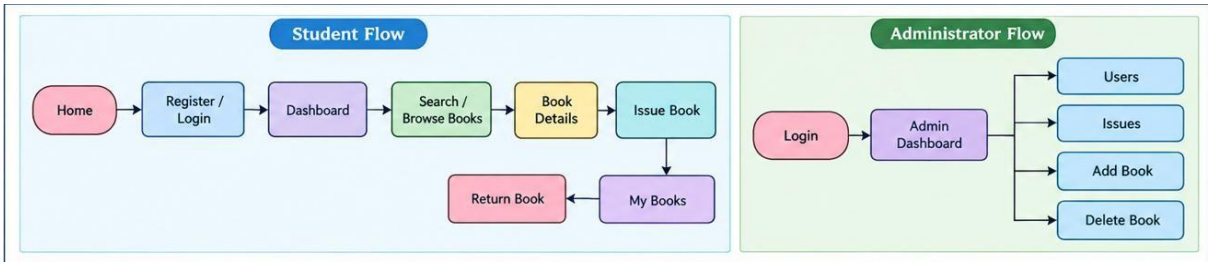
The uploaded database snapshot contains **7** user records, **964** book records, and **13** issue records. This is a snapshot of the supplied project database, not a performance benchmark.

⭘ SYSTEM DESIGN

1.Technical Architecture

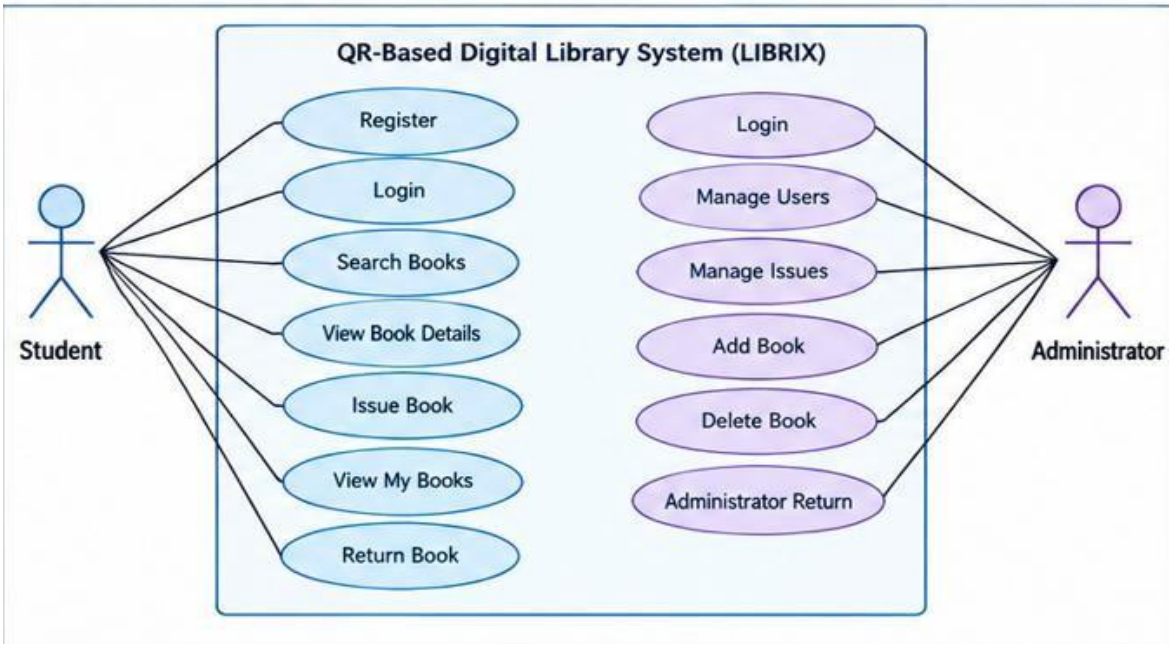
The application follows a simple web-application flow. A browser sends a request to Flask. Flask routes process authentication, book operations, and transactions. SQLAlchemy communicates with SQLite, while templates produce web pages. The QR-code component creates image files for book records.

2. System Flow Chart Student flow: **Home → Register/Login → Dashboard → Search/Browse Books → Book Details → Issue Book → My Books → Return Book**. Administrator flow: **Login → Admin Dashboard → Users / Issues / Add Book / Delete Book**



3. Use Case Diagram

The student actor can register, log in, search books, view details, issue books, view personal books, and return books. The Administrator actor can log in, manage users, manage issues, add books, delete eligible books, and perform administrator returns.



4. Data Flow / Database Relationship

User and Book are connected through Issue. A user may have multiple issue records, and a book may appear in multiple issue records over time. Issue stores dates and status, while Book stores quantity, available copies, shelf location, and QR-code path.

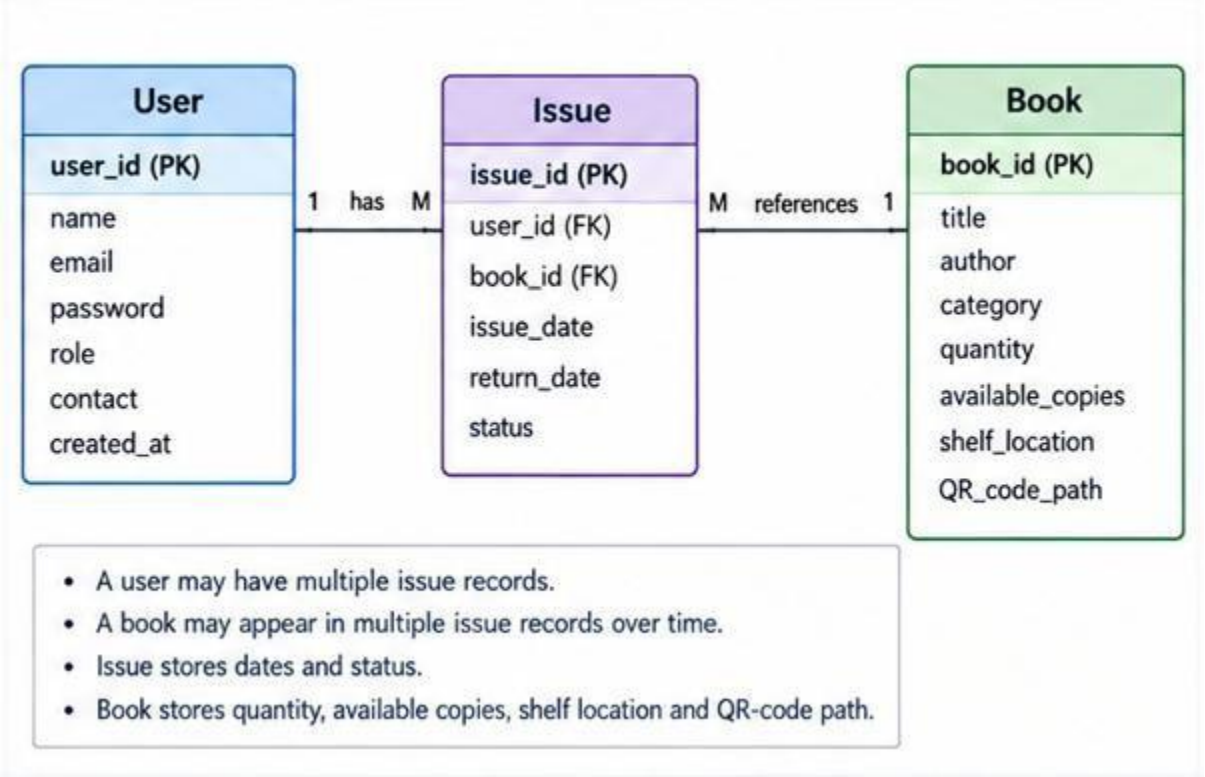


Fig 5.3 Data Flow / Database Relationship Diagram

IMPLEMENTATION

1. Modules Description

- Home Module: LIBRIX landing page and navigation.
- Authentication Module: registration, login, password verification, session creation, and logout.
- Dashboard Module: role-specific information and activity.
- Books Module: browsing, searching, suggestions, details, and availability filtering.
- Issue Module: creates issue transactions, decreases availability, and sets a 14-day due date.
- Return Module: records return date/status and increases availability.
- Admin Module: user, issue, and book management.
- QR Module: generates QR images and stores their paths.

2. Technology Used

The application is written in Python. Flask handles routing and requests. Flask-SQL Alchemy provides the ORM layer. SQLite stores data locally. Werk Zeug security functions handle password hashing and verification. The qrcode library creates book QR-code images. The supplied CSS files define LIBRIX home, authentication, dashboard, and responsive styling.

3. Database Tables

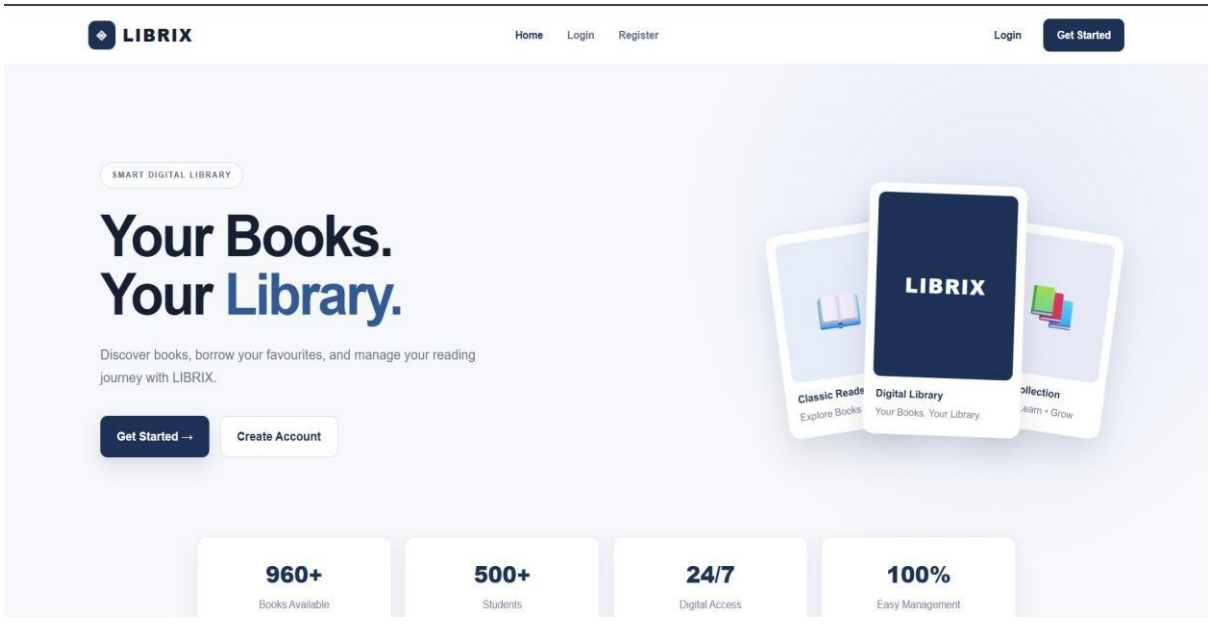
Table	Important Fields	Purpose
User	id, name, email, password, role	Users and access roles
Book	id, title, author, sign, category, quantity, available	shelf, rebook catalogue and QR information
Issue	id, user_id, book_id, issue_date, due_date, return	date, status Borrowing and return transactions

4. Important Implementation Rules

- Administrators cannot issue books through the student issue route.
- A book cannot be issued when no copy is available.
- A student cannot create a second active issue for the same book.
- The normal issue period is 14 days.
- A student can return an issue belonging to that student.
- An administrator cannot delete their own account or change their own role.
- A book cannot be deleted while it is currently issued.
- Adding a book validates required fields, quantity, and duplicate ISBNs.

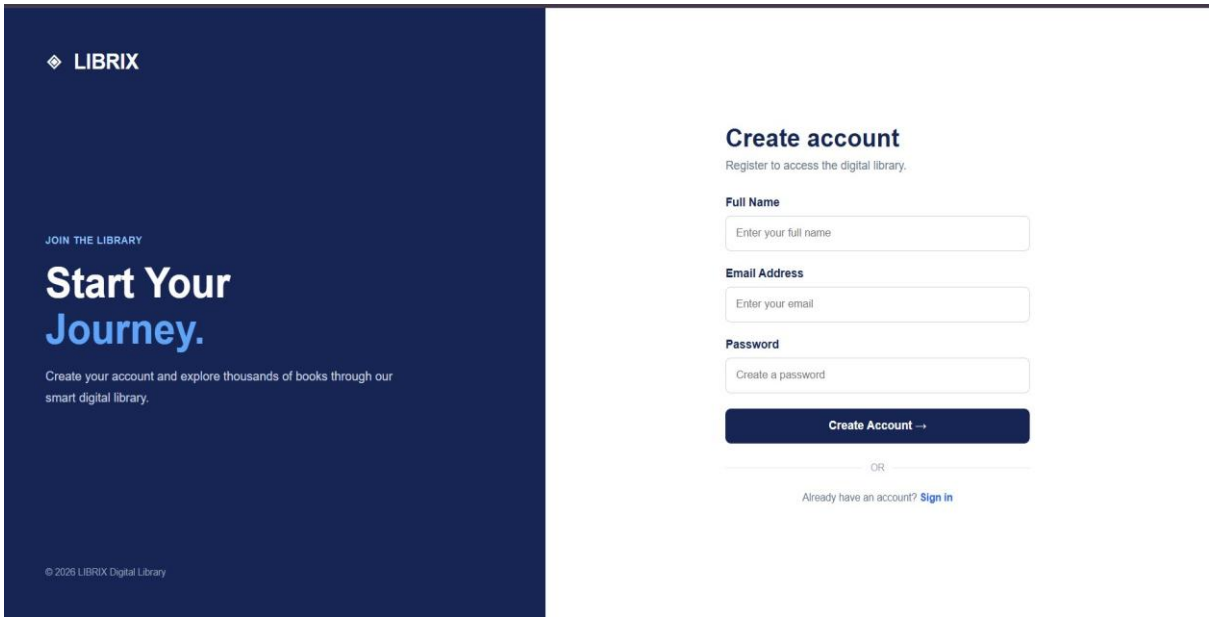
5. Screenshots / Output:

- **Home page:**

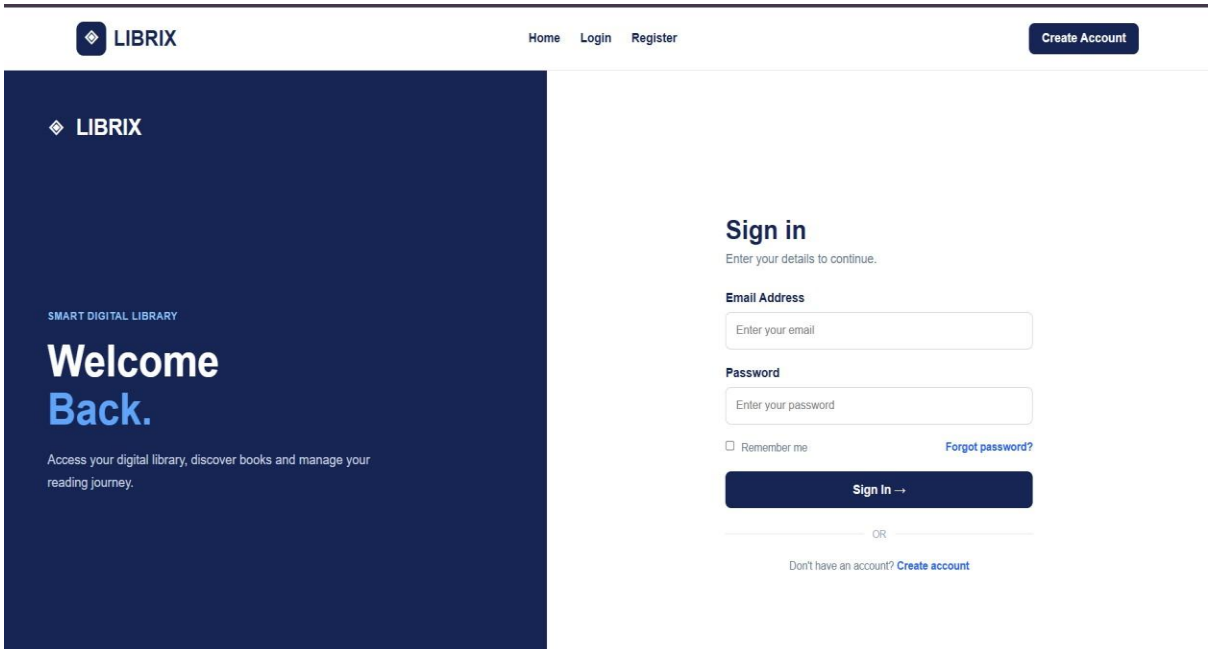


➢ Register Page / Login page:

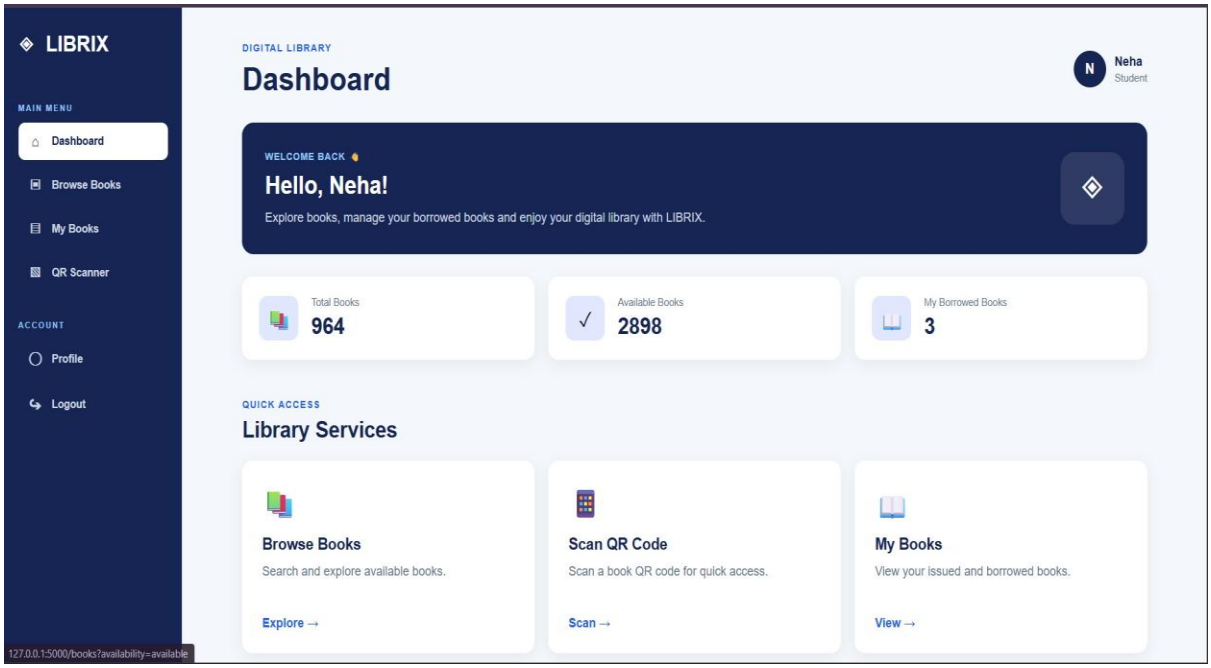
1. Register page



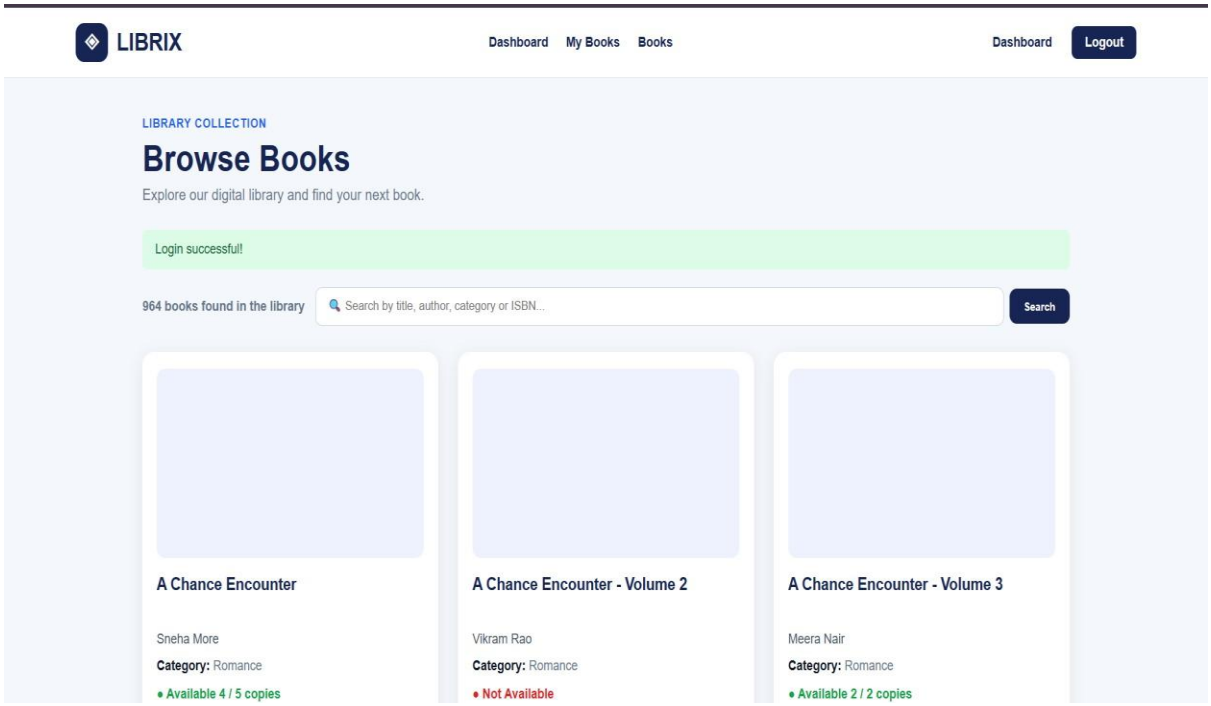
2. Login Page

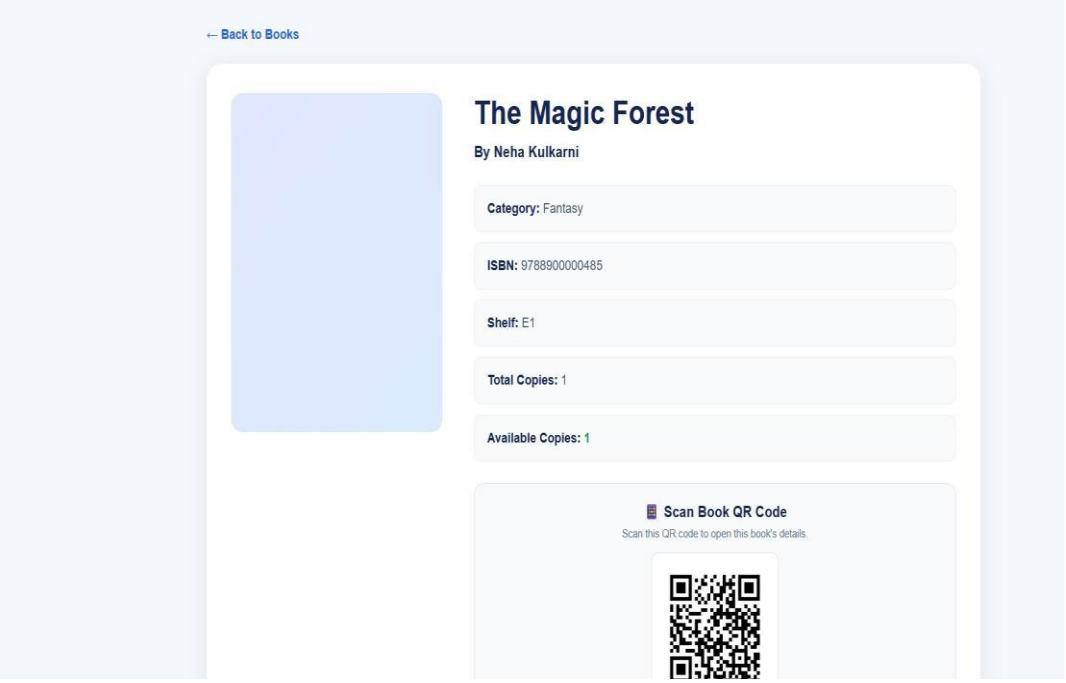


User Dashboard Page:

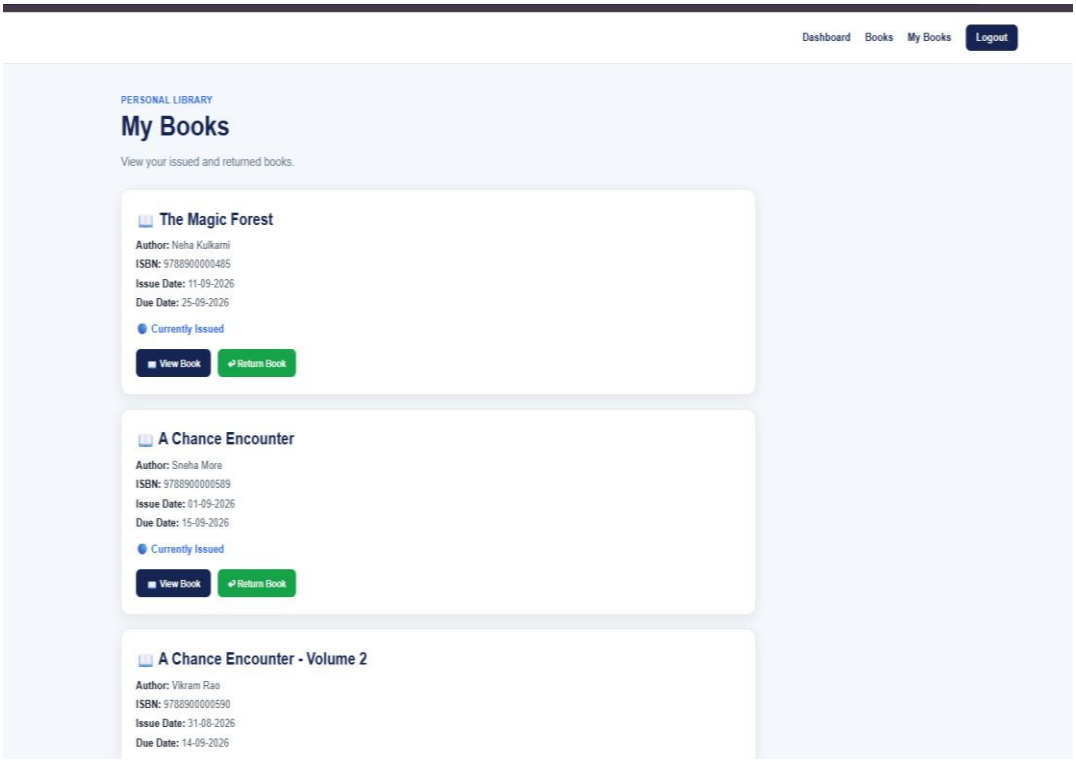


Browse / Search Book:

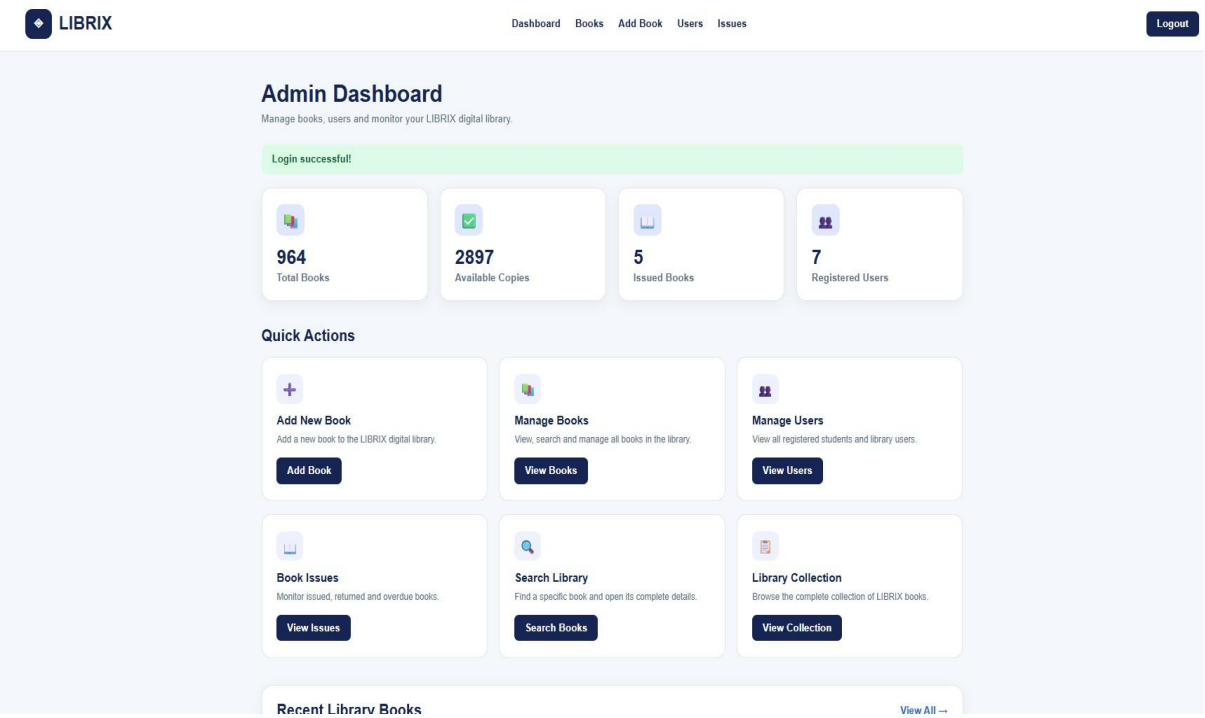



← Back to Books
The Magic Forest
By Neha Kulkarni
Category: Fantasy
ISBN: 9788900000485
Shelf: E1
Total Copies: 1
Available Copies: 1
Scan Book QR Code
Scan this QR code to open this book's details.

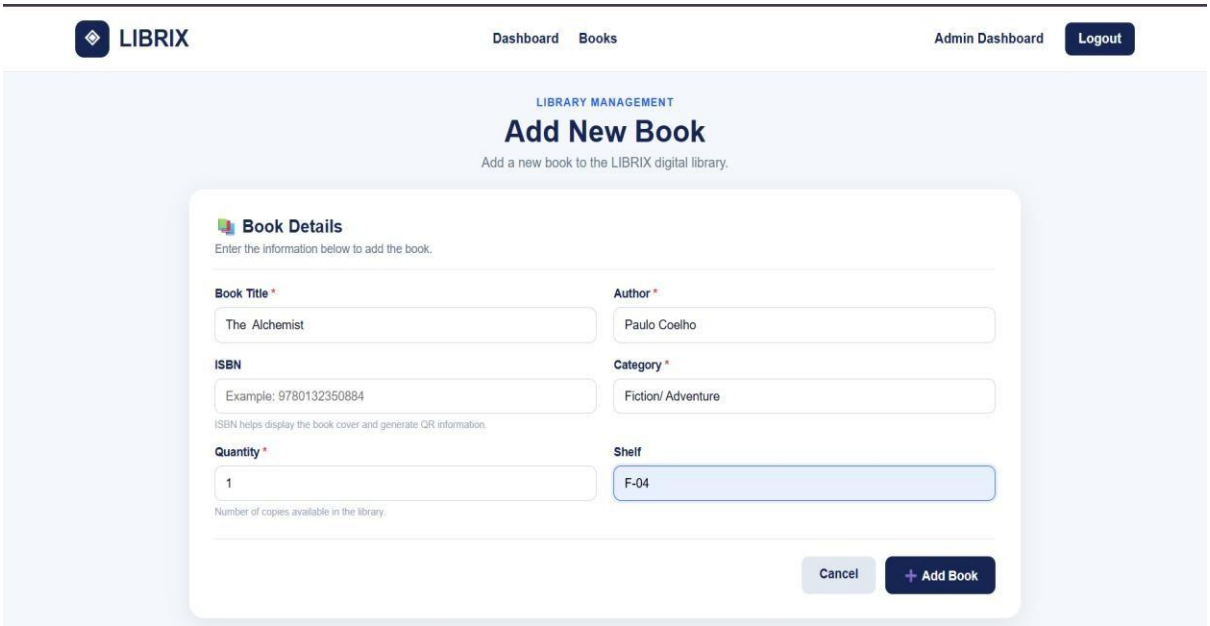
- **My Book**


Dashboard
Books
My Books
Logout
PERSONAL LIBRARY
My Books
View your issued and returned books.
The Magic Forest
Author: Neha Kulkarni
ISBN: 9788900000485
Issue Date: 11-09-2026
Due Date: 25-09-2026
Currently Issued
View Book
Return Book
A Chance Encounter
Author: Sneha More
ISBN: 9788900000589
Issue Date: 01-09-2026
Due Date: 15-09-2026
Currently Issued
View Book
Return Book
A Chance Encounter - Volume 2
Author: Vikram Rao
ISBN: 9788900000590
Issue Date: 31-08-2026
Due Date: 14-09-2026

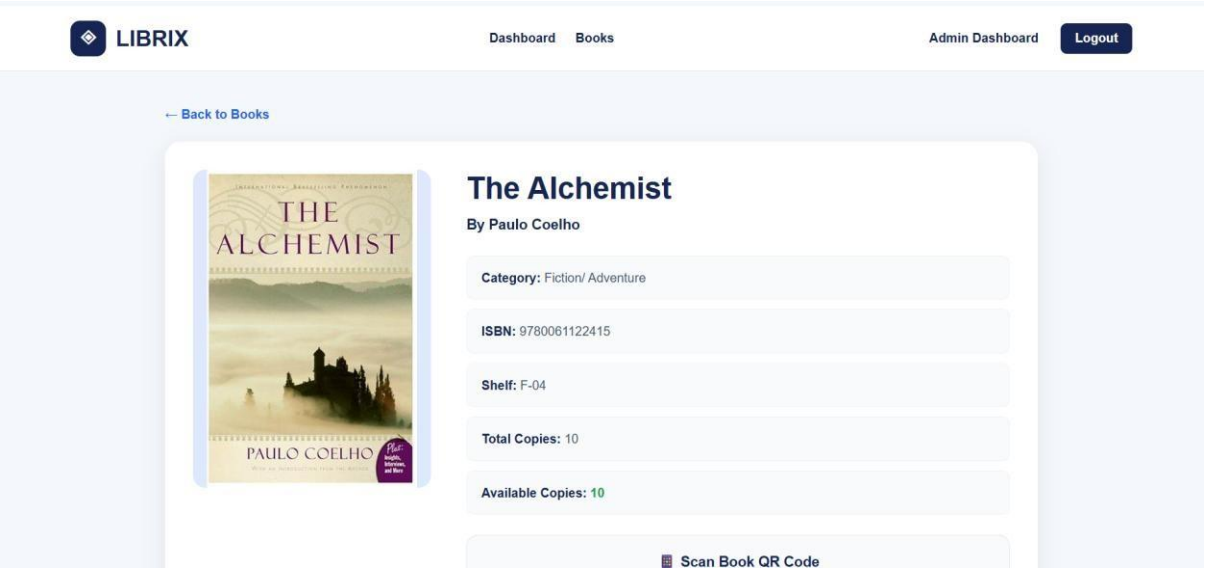
Admin Dashboard



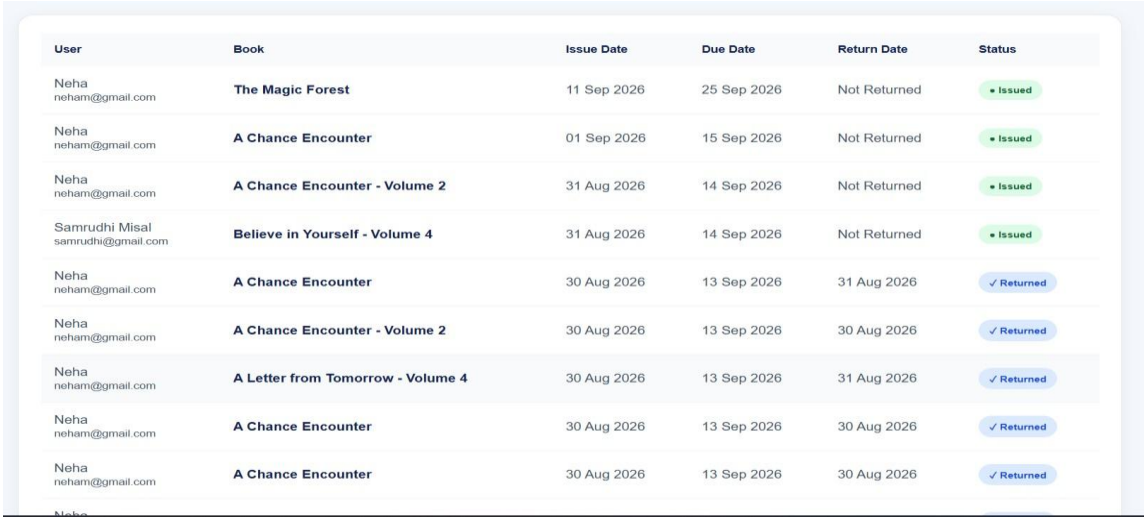
Add New Book



New book Added

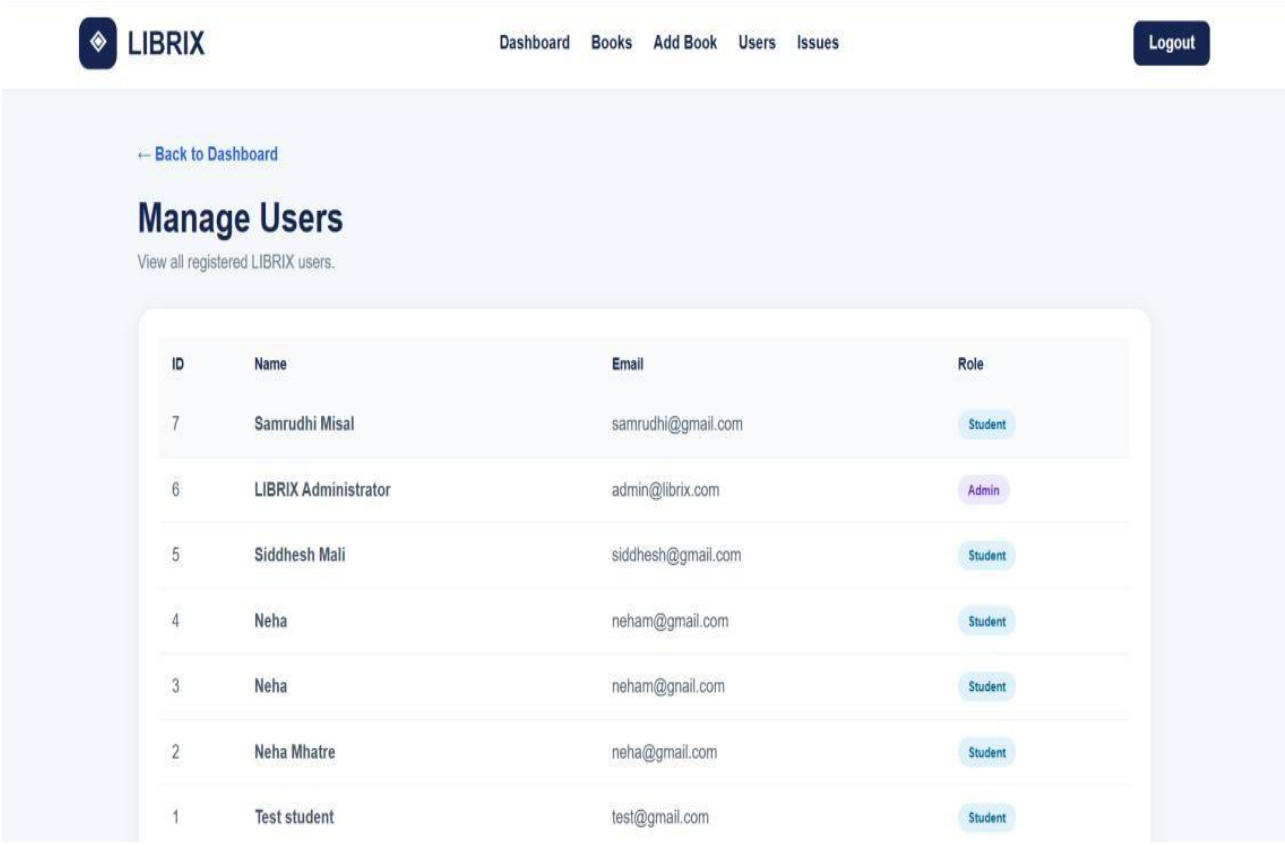


Issue / Return Books



User	Book	Issue Date	Due Date	Return Date	Status
Neha neham@gmail.com	The Magic Forest	11 Sep 2026	25 Sep 2026	Not Returned	• Issued
Neha neham@gmail.com	A Chance Encounter	01 Sep 2026	15 Sep 2026	Not Returned	• Issued
Neha neham@gmail.com	A Chance Encounter - Volume 2	31 Aug 2026	14 Sep 2026	Not Returned	• Issued
Samrudhi Misal samrudhi@gmail.com	Believe in Yourself - Volume 4	31 Aug 2026	14 Sep 2026	Not Returned	• Issued
Neha neham@gmail.com	A Chance Encounter	30 Aug 2026	13 Sep 2026	31 Aug 2026	✓ Returned
Neha neham@gmail.com	A Chance Encounter - Volume 2	30 Aug 2026	13 Sep 2026	30 Aug 2026	✓ Returned
Neha neham@gmail.com	A Letter from Tomorrow - Volume 4	30 Aug 2026	13 Sep 2026	31 Aug 2026	✓ Returned
Neha neham@gmail.com	A Chance Encounter	30 Aug 2026	13 Sep 2026	30 Aug 2026	✓ Returned
Neha neham@gmail.com	A Chance Encounter	30 Aug 2026	13 Sep 2026	30 Aug 2026	✓ Returned

View all registered LIBRIX users.



- **TESTING**

1. Testing Methodology

Testing content in this report is derived from the implemented application logic, supplied database, and project files. The full Flask application could not be launched in the current execution environment because Flask is not installed there. Therefore, this report does not falsely claim completed end-to-end runtime testing. The following are functional test cases to execute locally.

2. Test Cases

ID	Test Case	Expected Result
TC01	Valid registration	Account is created and user is redirected to login
TC02	Existing email registration	Registration is rejected
TC03	Valid login	Correct dashboard is opened
TC04	Invalid login	Login is rejected
TC05	Book search	Matching books are displayed
TC06	Issue available book	Issue is created, availability decreases, due dat 14 days later
TC07	Issue unavailable book	Issue is rejected
TC08	Duplicate active issue	Second issue is rejected
TC09	Return active issue	Status/date update and availability increases
TC10	Admin adds valid book	Book and QR-code path are created
TC11	Duplicate ISBN	Book is rejected
TC12	Delete issued book	Deletion is rejected
TC13	Unauthorized admin route	Access is denied/redirected
TC14	Logout	Session is cleared

CHAPTER 5:

RESULTS AND DISCUSSION

5.1 Implemented Results

Inspection of the supplied application shows routes for registration, login, dashboard access, book search, suggestions, book details, book issue, personal books, return, administrator dashboard, user management, issue management, administrator return, book addition, book deletion, and logout. The database schema confirms the three central entities required for the library workflow. The supplied database contains populated users, books, and issue records, demonstrating persistent SQLite storage. The supplied CSS contains dedicated styling for the LIBRIX home page, login/register screens, dashboard, statistics cards, categories, service cards, and responsive layouts.

5.2 Discussion

- Library records are centralized in a relational database.
- Role-based route protection separates student and administrator operations.
- Issue and return operations update book availability.
- Due dates are calculated automatically.
- QR-code generation gives each book a digital identification reference.
- Responsive CSS supports different screen sizes.

CHAPTER 6:

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The QR-Based Digital Library System (LIBRIX) provides a practical digital approach to common library operations. The supplied project implements authentication, book catalogue management, search, book issue and return processing, administrator controls, SQLite database storage, and QR-code generation. Users, books, and transactions are stored in separate but related tables.

The project demonstrates web development, database management, authentication, role-based access control, CRUD operations, transaction handling, and QR-code generation. The main limitation identified from the supplied files is that QR generation is implemented, while a complete physical QR-scanning workflow is not evident in the application code inspected for this report.

6.2 Future Scope

- Add a camera-based QR scanner.
- Add automatic fine calculation for overdue books.
- Add email or notification reminders.
- Add librarian reports and analytics.
- Add richer book information and cover images.

- Add pagination for large collections.
- Move from local SQLite to a hosted database for multi-user deployment.
- Add stronger production security and session controls.
- Add database backup and restore.
- Deploy the application to a server.

CHAPTER 7

REFERENCES, GLOSSARY & APPENDICES

• REFERENCES

- **Python documentation** — Python language and standard library concepts.
- **Flask documentation** — web application routing and sessions.
- **Flask-SQL Alchemy documentation** — SQL Alchemy integration with Flask.
- **SQLite documentation** — relational database storage.
- **Werk Zeug documentation** — password hashing and verification.
- **Python QR-code package documentation** — QR-code generation.
- Uploaded LIBRIX project files: application bytecode, models bytecode, seed-book bytecode, SQLite database, and CSS files.

• GLOSSARY

- **LIBRIX** – Name used by the project interface for the digital library system.
- **QR Code** – Two-dimensional code used to represent digital information.

- **CRUD** – Create, Read, Update, and Delete operations.
- **Flask** – Python web framework used by the project.
- **SQL Alchemy** – Database toolkit/ORM used through Flask-SQL Alchemy.
- **SQLite** – Lightweight relational database used by the project.
- **ORM** – Object-Relational Mapping, used to work with database records through application objects.
- **Session** – Application state used to retain logged-in user information across requests.
- **ISBN** – International Standard Book Number used to identify a book edition.
- **Issue** – A borrowing transaction connecting a user and a book.
- **Due Date** – Date by which an issued book is expected to be returned.
- **Role** – Access category such as student or administrator.

• APPENDICES

Appendix A: Database Structure

The supplied database confirms the main tables **user**, **book**, and **issue**. The Book table includes title, author, ISBN, category, quantity, available copies, shelf, and QR-code path. The Issue table includes foreign keys to User and Book plus issue, due, and return dates and a status value.

Appendix B: Sample Book Data

Book	Author	Category	Shelf
Introduction to Mathematics	Aarav Mehta	Study	A1
Introduction to Mathematics - Volume 2	Ananya Sharma	Study	B1
Introduction to Mathematics - Volume 3	Riya Patil	Study	C1
Introduction to Mathematics - Volume 4	Rahul Deshmukh	Study	D1

Fundamentals of Physics	Neha Kulkarni	Study	E1
Fundamentals of Physics - Volume 2	Aditya Joshi	Study	F1
Fundamentals of Physics - Volume 3	Priya Shah	Study	G1
Fundamentals of Physics - Volume 4	Karan Verma	Study	H1